Greenhouse Gas Footprints and Sectoral Eco Efficiency in Pakistan
A domestic input–output and undesirable-output SBM analysis

This study domestic GHG emissions in Pakistan caused by supply chains and assesses the eco-efficiency of 14 broad sectors. For these sectors, $MtCO_2e$ raised upstream for around 55.5% challenging of the industries supplying final demand. Among all these, exports are recorded as the particular factor of supply-chain dependence, under this domestic footprint accounts 77.2% of their increase. Manufacturing, utilities and transport remained below the frontier under all tested specifications. Trade, education and health passed a separate screen for lower embodied emissions and above-median employment supported per dollar of demand.

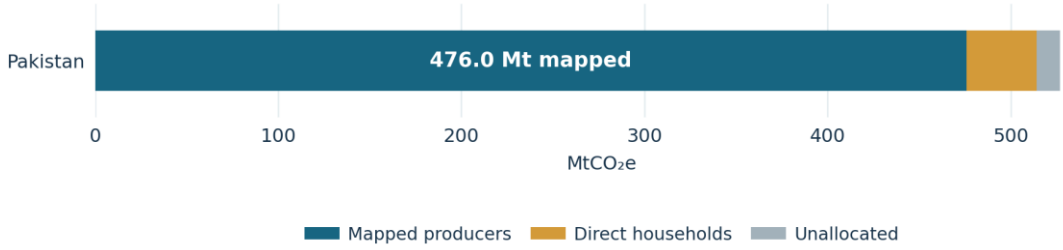


Figure 1. Coverage of Pakistan's greenhouse gas inventory

Emissions Embodied in Final Demand

Household consumption drove 371.9 $MtCO_2e$ of domestic producer emissions, or 78.1% of the mapped total. Another 38.3 Mt was assigned directly to households, yielding an estimated 410.2 Mt for the domestic household footprint on this boundary. Exports accounted for 42.4 Mt of producer emissions and capital formation for 30.8 Mt.

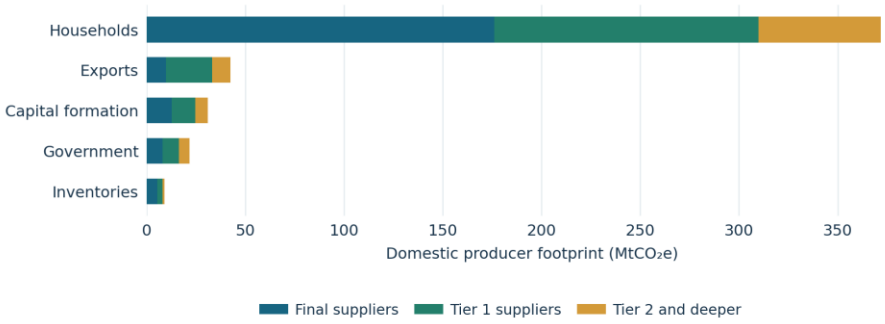


Figure 2. Domestic producer emissions by final-demand category and tier

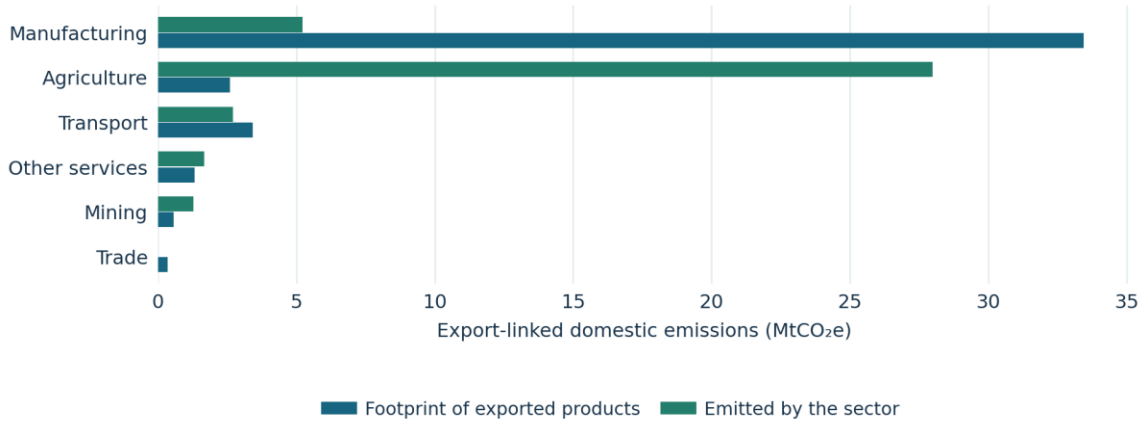


Figure 3. Exported-product footprints and the domestic industries emitting for exports

Manufactured exports carried 33.4 $MtCO_2e$, or 78.8% of the export footprint. Agriculture emitted 28.0 Mt in producing inputs for all exports, while agriculture's own exported products carried 2.6 Mt. This gap shows how an export footprint can be concentrated in manufacturing while the emissions themselves are concentrated in agriculture.

Supply Chain Hotspots and Sectoral Efficiency

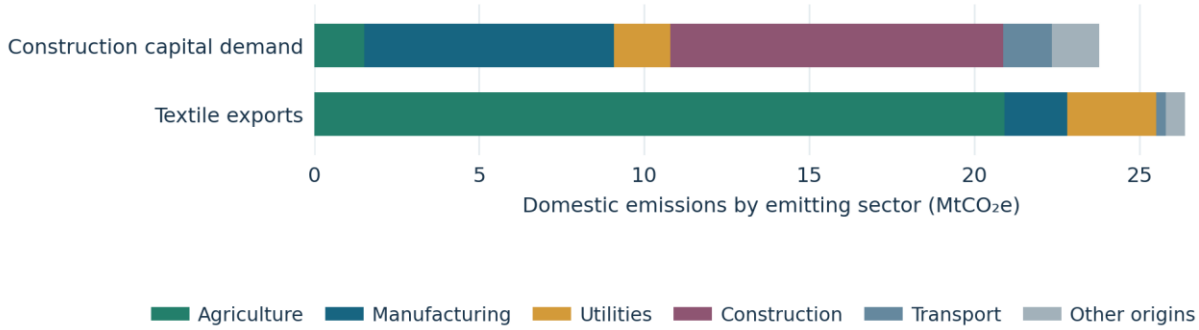


Figure 4. Emitting industries behind textile exports and construction capital demand

Textile exports embodied 26.4 $MtCO_2e$, including 20.9 Mt from agriculture and 2.7 Mt from utilities. Construction products bought for fixed capital formation embodied 23.8 Mt. Manufacturing suppliers contributed 7.6 Mt of that total, in addition to 10.1 Mt from construction itself. These results support examining agricultural suppliers in textile procurement and materials suppliers in infrastructure procurement.

Eco Efficiency Frontier

The non-oriented SBM model treats investment, employment and spending on refined fuels and utility supply as inputs. Gross value added is the desirable output and direct GHG is the undesirable output. The score divides one minus average proportional input excess by one plus average

proportional GVA shortfall and GHG excess. Variable returns to scale (VRS) is the baseline, with constant returns to scale (CRS) as a sensitivity case.

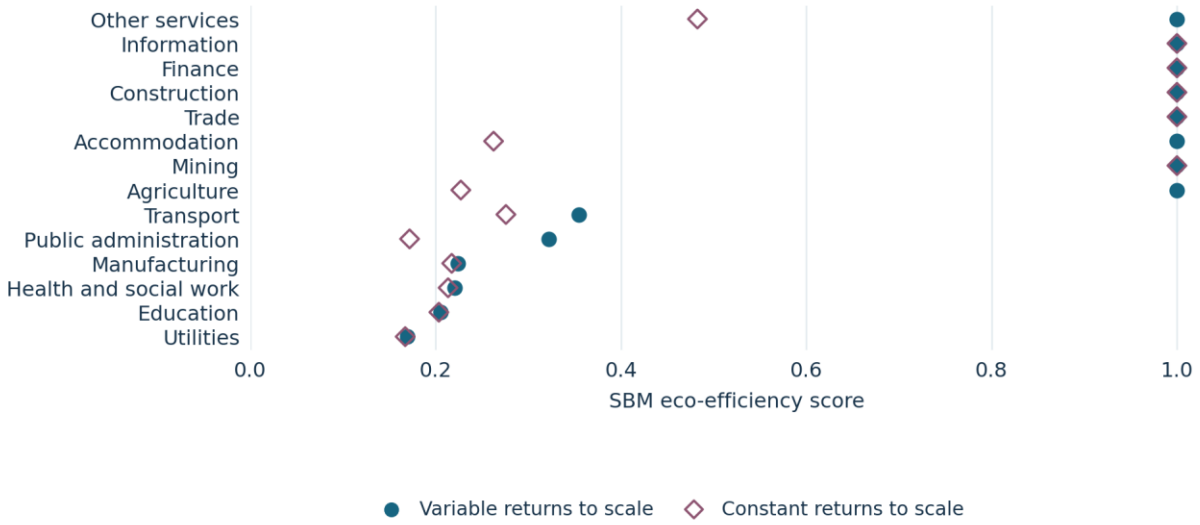


Figure 5. SBM eco-efficiency under variable and constant returns to scale

Utilities scored 0.170, manufacturing 0.224 and transport 0.354 under VRS. Agriculture was on the VRS frontier despite 242.3 $MtCO_2e$ of direct emissions. Its CRS score fell to 0.227. The difference illustrates how scale assumptions and the available peer set affect the benchmark.

Robustness and Employment Implications

Eight of 14 groups tied at the VRS frontier, compared with five under CRS. This limits discrimination across sectors with very different production processes. The spending proxy includes water and uses monetary flows in place of physical energy or a renewable split. GFCF measures annual investment. Sectoral HDI is unavailable in the matched sources. Geographical HDI measures health, education and living standards by place.

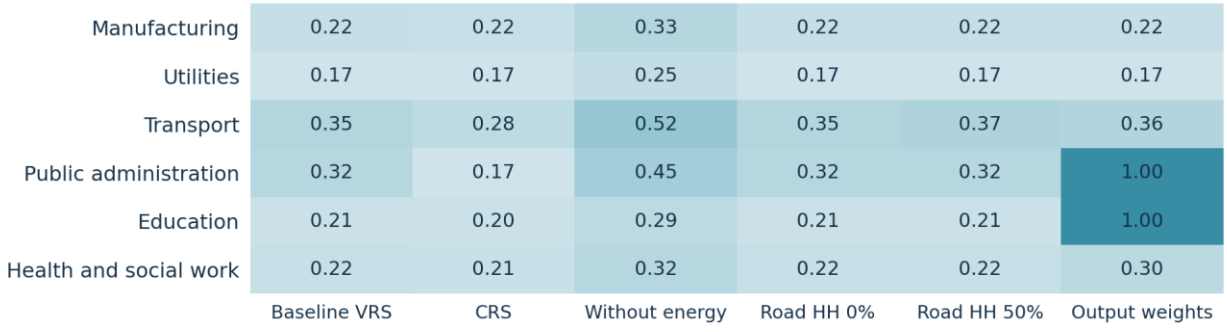


Figure 6. Sensitivity of the six baseline off-frontier sectors

Manufacturing, utilities and transport remain off the frontier in every specification shown. With output-based emission allocation, education moves from 0.205 to 1.000 and public administration from 0.322 to 1.000. Transport varies from 0.346 to 0.373 under the two alternative household road shares. These tests distinguish robust off-frontier findings from scores driven by the source crosswalk.

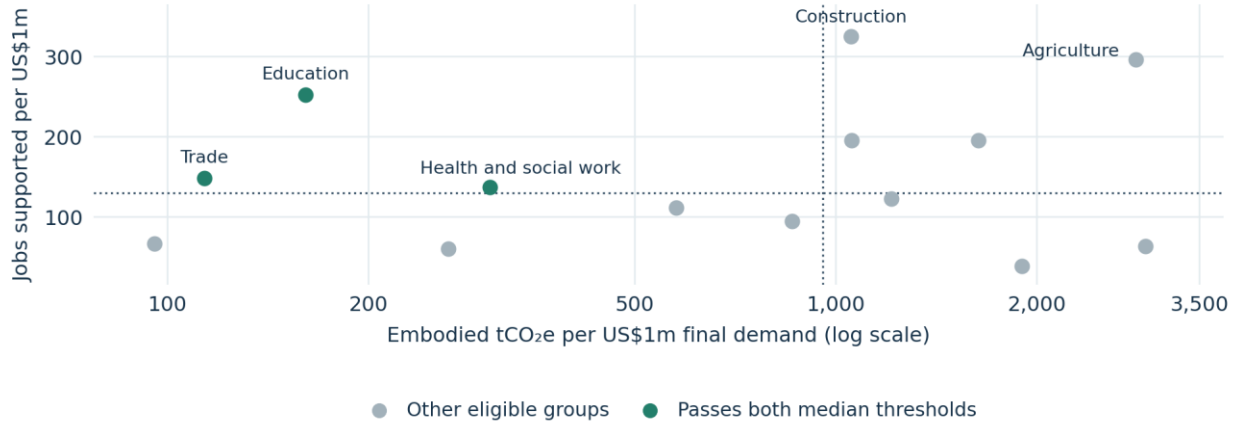


Figure 7. Employment and embodied emissions per US$1 million of final demand

Trade, education and health meet both screening thresholds of at least 129.7 jobs supported and at most 956.8 tonnes CO_2e per US$1 million of final demand. These multipliers hold labour productivity fixed. They describe employment supported at benchmark rates and leave job quality, distributional effects and welfare gains unmeasured.

Investment Screening and Conclusions

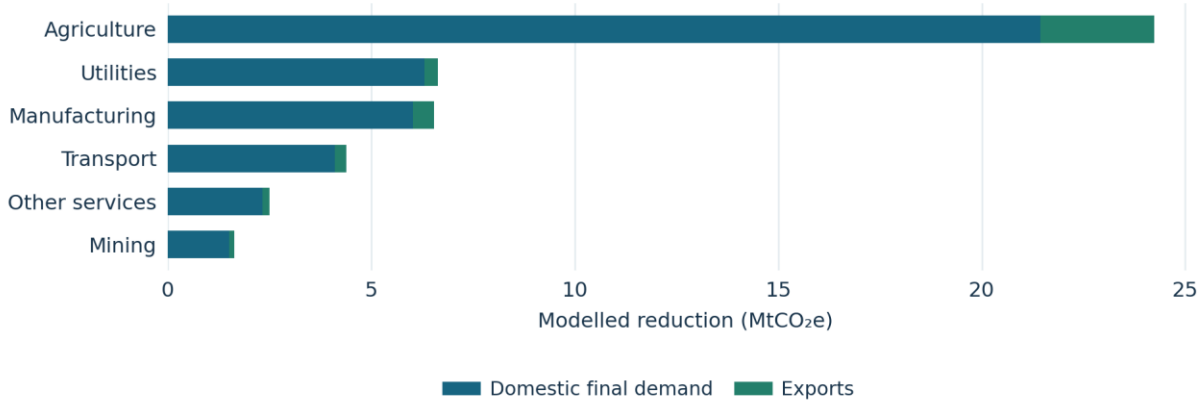


Figure 8. Domestic footprint reduction under a 10% source-intensity cut

A 10% intensity cut in agriculture would reduce the benchmark producer footprint by 24.2 $MtCO_2e$, including 2.8 Mt linked to exports. Applying the same cut separately to utilities or manufacturing would reduce it by 6.6 Mt or 6.5 Mt, respectively. The large agriculture result reflects its emissions base. Costs and technical feasibility must be evaluated for each intervention.